

CASE STUDY 7: ICON WATER - EXERCISING GENERAL ENVIRONMENTAL DUTY IN THE ACT



Icon Water has applied risk-based decision-making tools during liaison with environmental regulators to identify appropriate, feasible and prudent approaches to managing wet weather overflows.

Management approach

Risk-based

Receiving environment principles considered/met/achieved

Ecosystem, Community

Background

Icon Water has a general environmental duty to take reasonable and practicable steps to minimise environmental harm or nuisance from activities under Part 3 Section 22 of the ACT Environment Protection Act 1997 (EP Act). Compliance with the duty allows for consideration of the environment and planning context, current technical knowledge, financial implications, the likelihood and degree of success of harm minimisation and any other relevant circumstances.

Investigations and Program Objectives

The Canberra Sewer Network is designed to contain sewage flows associated with up to a 1 in 10 year ARI event standard consistent with the approved Environment Management Plan for the Canberra Sewer Network. There are ~60 designed overflow points. Performance may not always adhere to design standards due to incremental growth in the sewage catchment beyond pipe/pump capacity, failures in power supply to sewer pump stations or climate change with more extreme weather events.

Icon Water developed a risk-based approach to compare potential options using decision-making criteria based on the provisions of the EP Act to demonstrate compliance with the general environmental duty and to support optimal liaison with the environmental regulator. A spreadsheet was developed at asset level outlining baseline information for a defined wet weather overflow event, including:

- infrastructure performance – design standard requirement, average dry weather flow, peak wet weather flow, retention volumes
- site environment and planning context – land use, sensitive receivers, planning layers, water catchment code, protected matters

Stakeholder Involvement

A series of options were identified through workshops with asset managers, operators and environmental scientists as potential solutions to address the event risk. These options ranged from 'do nothing' (which may already include monitoring, alarms and operational controls), through to reduction in inflow and infiltration, screens, new valves, upsizing of pipes/pumps, overflow storage containment and new or renewed sewer mains. Preferred options were then identified using multi-criteria evaluation based on:

- the nature and sensitivity of the receiving environment – description and spatial imagery

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- the regulatory compliance, reputation and environmental risks of the event (with the option controls in place)
- current technical knowledge - asset condition, configuration, feasibility, performance
- financial implications - both capital and operational cost including for incident response
- the extent to which the option is likely to minimise the environmental harm (measured by assessing the likelihood and degree of success of the option) - subject to technical feasibility of option, typically the inverse of risk of the event with option controls in place
- any other relevant circumstances - growth modelling, geotechnical uncertainty, manual versus automated control, construction disruption, land management constraints, implicit benefits/costs

Outcomes

The process to apply an environmental risk-based assessment of options was demonstrated to the environmental regulator. The environmental regulator provided feedback that they were confident in the risk based decision-making framework as it clearly applies the tests required by the EP Act. Solutions were identified, discussed and agreed with the environmental regulator at individual asset level only. Both internal and external stakeholders found the risk-based approach useful in identifying, evaluating and selecting appropriate, feasible and prudent options for managing wet weather overflows. The method has facilitated consistent synthesis of both qualitative and quantitative information and demonstrated compliance with the general environmental duty.

Costs/benefits

The use of this risk based decision-making framework for one capital works program resulted in savings of ~\$173m by defining the benefits of a combined operational and capital solution at a cost of ~\$33m as opposed to the previously preferred option of upsizing the existing sewer main to for wet weather overflow design standard at a cost of ~\$200m (with a margin of error of +/- 70%).

The value of overall savings at a program level has not been quantified. The approach has justified the benefits of applying a risk-based approach to determine investment requirements on a case by case basis, as opposed to applying standard major capital improvements across a program scale (to manage wet weather overflows).

Project Timing

Environmental duty risk assessments continue to be undertaken for specific assets and locations as part of asset planning and for the capital expenditure program.



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